

Artificial Intelligence and RAG Systems Overview

1. Introduction to AI and Machine Learning

Artificial Intelligence (AI) is a field of computer science dedicated to solving cognitive problems commonly associated with human intelligence, such as learning, problem solving, and pattern recognition.

2. Natural Language Processing (NLP)

NLP allows computers to understand and respond to text or voice data. Modern NLP relies heavily on Transformer architectures like those used in Gemini models to understand the semantic meaning of human speech.

3. Retrieval-Augmented Generation (RAG)

RAG is a technique that optimizes the output of a Large Language Model. It references an authoritative knowledge base outside of its training data sources before generating a response, usually by utilizing Vector Embeddings and Cosine Similarity to search a database.